

Linear Algebra Notes

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Fall 2025

These notes are based on the course *Introduction to Linear Algebra (Math4018)* taught by Prof. Kowk-Wing Tsoi at the Department of Mathematics, NTU. The structure follows the material prepared by Prof. Tsoi, with some adjustments and simplifications to aid exercises and comprehension. The details and complete arguments are documented in my handwritten notes.

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Contents

1	Fields and Vector Space	3
2	Basis and Dimension	6
3	Linear Transformation	11
3.1	Linear function	11
3.2	Matrix Operations	14
3.3	Change of coordinate & Linear Operator	18
4	Determinants	20
5	Diagonalization	25
5.1	Similarity	25
5.2	Diagonalizability	27
5.3	Application of Diagonalization	32
6	Linear Duality	35
7	Jordan Canonical Form	40
8	Inner Product Space	48

1 Fields and Vector Space

Fields generalize the concepts of scalars. A set F is called a field if we can define addition and multiplication operations to satisfy 11 axioms.

Exercise 1.1 .

Write down the 11 axioms (5+5+1). These rules ensure we can add, subtract, multiply and divide by non-zero element in the field.

Example 1.1 .

Some non-examples of fields:

1. $\mathbb{Z}$ is not a field (division).
2. $M_n(\mathbb{R})$ is not a field (commutativity under multiplication).

A set V is called a vector space over a field F if the space endowed with two operations satisfies 10 axioms.

Exercise 1.2 .

Write down the 10 axioms (5+5). These rules help us determine whether a given space is a vector space.

Example 1.2 .

Let F be a field ($\mathbb{R}, \mathbb{C}, \mathbb{Q}, F_2$). $F^n = \{(x_1, x_2, \dots, x_n)\}$ is a vector space over F . In general, if two fields $F_1 \subseteq F_2$, then F_2 is a vector space over F_1 .

Exercise 1.3 .

Prove for each $\mathbf{v} \in V$, its additive inverse $-\mathbf{v}$ is unique.

Proof.

Suppose $\exists \mathbf{v}' \neq \mathbf{v}''$ be two additive inverse of $\mathbf{v}$. Use $\mathbf{0}$ to relate $\mathbf{v}' = \mathbf{v}''$. ■

Use the axioms to derive the related property:

Property 1.1 .

Let V be a vector space over F . Let $\mathbf{v} \in V$ and $a \in F$.

1. $0 \cdot \mathbf{v} = \mathbf{0}$
2. $a \cdot \mathbf{0} = \mathbf{0}$
3. $(-a)\mathbf{v} = -(a\mathbf{v})$
4. $a(-\mathbf{v}) = -(a\mathbf{v})$

Proof.

The first two: let LHS be a vector $\mathbf{w}$, and use $\mathbf{w} + \mathbf{w} = \mathbf{w}$ to show $\mathbf{w} = \mathbf{0}$. The last two: use the uniqueness of additive inverse and the result of 1, 2. ■

Definition 1.1 .

Let V be a vector space over F . $W \subseteq V$ is a subspace of V if W is also a vector space over F with addition and scalar multiplication inherited from V .

Exercise 1.4 .

Write down the sufficient and necessary condition of a subspace.

Theorem 1.1 (Subspace of $\mathbb{R}^n$).

Let A be a m by n matrix. Then the solution space of a homogeneous linear system of equations:

$$W = \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}$$

forms a subspace of $\mathbb{R}^n$.

Proof.

We can prove W is a subspace directly using the above theorem. Alternatively, consider a linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the matrix A is exactly the matrix representing f . By construction, W is the kernel of f . $\ker(f)$ is a subspace of domain $\mathbb{R}^n$ (We will prove this in the future). ■

Theorem 1.2 .

Fix a positive integer n . The subset of trace-free matrices:

$$W = \{A \in M_n(F) : \text{tr}(A) = 0\}$$

forms a subspace of $M_n(F)$.

Proof.

This proof is similar to the previous one and left as an exercise. ■

Example 1.3 .

Differentiable function space is a subspace of continuous function space. This is an easy exercise.

Theorem 1.3 .

Intersection of subspaces is also a subspace.

Proof.

Left as an exercise. ■

Example 1.4 .

Consider a vector space $M_n(F)$, W_1 be a symmetric matrices, and W_2 be a upper-triangular matrices. The intersection $W_1 \cap W_2$ is a diagonal matrices, which is a subspace of the n -square matrices vector space.

Lastly, we introduce some concepts earlier.

Property 1.2 .

Reminder:

- 1. Union of two subspaces is not necessarily a subspace.*
- 2. Intersection of two subspaces is a subspace.*
- 3. The sum of two subspaces is the smallest subspace containing both.*
- 4. Spanning set is a subspace.*

2 Basis and Dimension

Let V be a (finite-dimension) vector space over a field F . Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a finite subset of V .

Definition 2.1 (Linear Combination).

Any vector of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n \text{ where } a_i \in F$$

is called a linear combination of S .

Definition 2.2 (Linear Independence).

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \mathbf{0} \text{ implies } a_i = 0, i = 1, 2, \dots, n$$

The only way to express the zero vector as a linear combination of S is to take all scalars to zeros.

S is linear dependent if it is not linearly independent.

Theorem 2.1 .

S is linearly dependent iff either $\mathbf{v}_1 = \mathbf{0}$ or for some r , $\mathbf{v}_r$ is a linear combination of the preceding vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$.

Proof.

($\Rightarrow$) Let r be the largest integer such that $a_r \neq 0$.

($\Leftarrow$) There exists a linear combination and show that the scalars are not all zeors. ■

Definition 2.3 (Linear Span).

The set of all linear combinations of S is:

$$\text{span}(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n : a_i \in F\}.$$

Note that $\text{span}(\emptyset) = \{\mathbf{0}\}$.

Theorem 2.2 .

$\text{span}(S)$ is a subspace of V .

Definition 2.4 (Spanning / Generating Set).

We say that S spans V / S is a spanning/generating set of V if $\text{span}(S) = V$.

For every $\mathbf{v} \in V$ can be expressed as a linear combination of S .

Definition 2.5 (Basis).

A subset S of V is a basis of V if

1. S is linearly independent.
2. S spans V .

If S is a finite set, then V is finite-dimensional. Otherwise, V is infinite-dimensional. Additionally, a basis is not unique.

Remark .

If S spans V and $\mathbf{w} \in S$, then $\{\mathbf{w}\} \cup S$ is not linearly independent. Conversely, if $\mathbf{w} \notin S$, then $\{\mathbf{w}\} \cup S$ is still linearly independent.

Theorem 2.3 .

Suppose S is a basis of V . Then for every $\mathbf{v} \in V$, there exists a unique collection of scalars $a_1, a_2, \dots, a_n$ such that

$$\mathbf{v} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n.$$

Proof.

When we want to show the uniqueness, we could prove by contradiction. That is, fix a vector which could be expressed as two different linear combinations of S . ■

Definition 2.6 (Coordinates of vector).

Fix an order of elements in a basis S . We call $(a_1, a_2, \dots, a_n)$ is the coordinates of $\mathbf{v}$ w.r.t the basis S .

The coordinates are denoted by $[\mathbf{v}]_S = (a_1, a_2, \dots, a_n)$, and the same vector could have different coordinates w.r.t different bases or different orders of the same basis.

Definition 2.7 (Dimension).

Suppose V is finite-dimensional vector space. The dimension of V is the size of a /any basis of V .

$$\dim_F(V) = |S|,$$

where S is a basis of V , $|\cdot|$ represents the size / cardinality of a set.

Note that $\text{span}(\emptyset) = \{\mathbf{0}\}$. The empty set spans zero vector space, and the empty set forms a basis of $\{\mathbf{0}\}$, $\dim_F(\{\mathbf{0}\}) = |\emptyset| = 0$. So zero vector space is the only zero-dimensional vector space.

Theorem 2.4 (Exchange / Replacement Theorem).

Suppose $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a spanning set of V and $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ is linearly independent. Then $m \leq n$.

Theorem 2.5 .

Let V be a vector space over F and W be a subspace of V . Then $\dim_F(W) \leq \dim_F(V)$.

Proof.

If V is infinite-dimensional, there is nothing to prove. Now suppose V is finite-dimensional. Claim: W must be finite-dimensional. Suppose, by contradiction, there exists a subspace with an infinite basis B_W . B_W is linearly independent in V . Let B_V be a basis of V , which is a finite-dimensional spanning set in V . By replacement theorem, which is a contradiction. Note the above argument uses the axiom of choice. ■

Theorem 2.6 .

If both S_1 and S_2 are both bases of V , then $|S_1| = |S_2|$.

Proof.

Recall the definition of basis. Apply the exchange theorem to S_1 and S_2 from the 'both' sides. ■

Theorem 2.7 .

Let V be a vector space of dimension n and $S \subseteq V$ be a subset.

1. If $|S| > n$, then S is linearly dependent.
2. If $|S| < n$, then S does not span V .

Proof.

1. Suppose, by contradiction, that S is linearly independent. Let B be any basis of V . We have $|B| = n$. By the exchange theorem, $|S| \leq |B| = n$, which is a contradiction. Thus S is linearly dependent.
2. Suppose, by contradiction, that S spans V . Let B be any basis of V . We have $|B| = n$. B is linearly independent and S spans V , by the exchange theorem, $|B| = n \leq |S|$, this contradicts $|S| < n$.

■

The proof from the above theorem provides a refined version of the exchange theorem:

Theorem 2.8 (Refined Exchange / Replacement Theorem).

Let T be any linearly independent subset of V . Let B be any basis of V . Let S be any spanning set of V . Then $|T| \leq |B| \leq |S|$, where $|B| = \dim_F(V)$.

Theorem 2.9 .

Let V and W be two finite-dimensional v.s. If $W \subseteq V$ and $\dim_F(W) = \dim_F(V)$, then $V = W$.

Proof.

It suffices to show that $W \supseteq V$. Suppose by contradiction, ... Let S be a basis of W . Then we have $W = \text{span}(S)$ and $\mathbf{v} \notin \text{span}(S)$. Construct the larger size of linearly independent set in V . Apply the replacement theorem, then we have a contradiction. ■

Theorem 2.10 .

Let V be a v.s. of dimension n . Then any n linearly independent subset of forms a basis of V .

Proof.

Easy proof. Let $W = \text{span}(S)$ and show that $W = V$. Recall any spanning set is a subspace of V . ■

Exercise 2.1 .

The above theorem shows that a maximal linearly independent set forms a basis of V . In duality, a minimal spanning set also forms a basis of V . Prove the theorem.

Theorem 2.11 .

Let $S = \{\mathbf{v}_1, \mathbf{v}_2 \dots \mathbf{v}_n, \mathbf{w}\}$. If $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then $\text{span}(S) = \text{span}(S \setminus \{\mathbf{w}\})$.

Apply (2.1) iteratively, we have the following sifting method.

Theorem 2.12 (Sifting method).

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. There exists $S' \subseteq S$ s.t. S' is linearly independent and $\text{span}(S') = \text{span}(S)$.

Theorem 2.13 (Basis Extension Theorem).

Let S be a linearly independent subset of V . Then S can be extended into a basis of V .

Proof.

Let B be any basis of V . Consider $S \cup B$. Apply the sifting method to $S \cup B$.

Question: Does S remain in the sifted set? ■

Theorem 2.14 (Duality of the Basis Extension Theorem).

Given S spans V , then there exists a subset $S' \subseteq S$ forms a basis of V .

Now let us prove replacement theorem

Proof.

Let $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a spanning set of V and $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be linearly independent. Idea is that insert each $\mathbf{w}_i$ in front of S_1 and sift. Think why $\{\mathbf{w}_i\} \cup S$ must be linearly dependent, then we get the inequality. ■

Here we introduce sum of spaces as a subspace.

Exercise 2.2 .

Let W_1 and W_2 be two subspaces of a vector space V .

1. *Write down the definition of sum of spaces.*
2. *Prove this is smallest subspace containing W_1 and W_2 .*
3. *Write down the definition of the direct sum of spaces.*
4. *Write down the dimension formula when the subspace is a direct sum of two spaces.*

Theorem 2.15 (Dimension Formula for Sum of Subspaces).

Let V_1 and V_2 be two subspaces of a finite-dimensional vector space V . Then

$$\dim_F(V_1 + V_2) = \dim_F(V_1) + \dim_F(V_2) - \dim_F(V_1 \cap V_2).$$

Proof.

Note that $W_1 \cap W_2 \subseteq W_1, W_2 \subseteq W_1 + W_2$. Start with the smallest set, apply basis extension theorem w.r.t W_1 and W_2 . Check the union of the three basis, S , forms the basis in $W_1 + W_2$, i.e. S is L.I. and spans $W_1 + W_2$. ■

3 Linear Transformation

3.1 Linear function

Let $f : V \rightarrow W$ be a linear transformation, and V is finite dimensional.

Definition 3.1 (linear function/transformation/map/homomorphism).

Check the function f : (i) Distribution on addition (ii) Scalars can be taken out of f .

Example 3.1 .

Verify the following examples are linear maps.

1. Differentiation function: $D : \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}[x]_{\leq n-1}$
2. Trace: $tr : M_n(F) \rightarrow F$
3. Fix a continuous function $g \in C(\mathbb{R})$. $S_g : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ defined by $S_g(f(x)) = f(g(x))$

Theorem 3.1 .

If $f : V \rightarrow W$ is a linear map. Then $f(\mathbf{0}_V) = \mathbf{0}_W$.

Alternatively, one can verify linearity via a contrapositive argument.

Remark .

Note the following difference:

1. $C(\mathbb{R}) \rightarrow C(\mathbb{R}) : f(x) \rightarrow f(x-1)$ is about replacing the domain with some function.
2. $\mathbb{R}^2 \rightarrow \mathbb{R}^2 : \{(x, y) : y = f(x)\} \rightarrow \{(x, y) : y = f(x-1)\}$ is about shifting the graph of function $f(x)$ to 1 unit to the right.

Theorem 3.2 .

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of V , then f is determined by $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)\}$, i.e.,

1. For every $\mathbf{v} \in V$, $f(\mathbf{v})$ is a linear combination of $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)\}$. Actually, this shows $\text{im}(f) = \text{span}(\{f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)\})$.
2. Suppose the linear map $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i) \forall i = 1, 2, \dots, n$. Then $g(\mathbf{v}) = f(\mathbf{v})$ for all $\mathbf{v} \in V$.

Corollary .

If f is an injection ($\ker(f) = 0$), then $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)\}$ forms a basis in $\text{im}(f)$.

The definitions of kernal and image are left as an exercise.

Theorem 3.3 .

Let $\dim(V) = n$ and $\dim(W) = k$

1. $\ker(f)$ is a subspace of V .
2. $\dim(\ker(f)) = \text{nullity}(f) = 0 \Leftrightarrow \ker(f) = \{\mathbf{0}_w\} \Leftrightarrow f$ is injective.
3. $\text{im}(f)$ is a subspace of W .
4. $\dim(\text{im}(f)) = \text{rank}(f) = k \Leftrightarrow \text{im}(f) = W \Leftrightarrow f$ is surjective.

Theorem 3.4 (Rank-Nullity Theorem).

$$\text{nullity}(f) + \text{rank}(f) = \dim_F(V)$$

Proof.

Start with the smallest set.

Since $\ker(f) \subseteq V$, let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$ and extend it to a basis of V , denoted by $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$.

Show that $\{f(\mathbf{w}_1), f(\mathbf{w}_2), \dots, f(\mathbf{w}_k)\}$ forms a basis of $\text{im}(f)$. ■

Theorem 3.5 .

If $f : V \rightarrow W$ is a linear bijection, then f^{-1} is also linear.

Note that if f is bijective, then the inverse function of f (f^{-1}) exists.

Definition 3.2 (Linear Isomorphism).

Let V and W be two finite dimension vector spaces over a field F .

1. $f : V \rightarrow W$ is a (linear) isomorphism from V onto W ($V \cong W$) if f is a bijective linear map.
2. V and W are said to be isomorphic as vector spaces if there exists a isomorphism $f : V \rightarrow W$.

An trivial but important example is that $\text{id}_V : V \rightarrow V$ is a isomorphism. So every vector space is isomorphic to itself.

Definition 3.3 (Homeomorphism).

Let (X, d_X) and (Y, d_Y) be two metric spaces.

1. $f : X \rightarrow Y$ is a homeomorphism ($X \cong Y$) if f is bijection, f is continuous

and f^{-1} is also continuous.

2. X and Y are said to be homeomorphic as vector spaces if there exists a homeomorphism $f : X \rightarrow Y$.

Theorem 3.6 .

Fix S as a basis of V , the coordinate mapping $\phi : V_S \rightarrow F^n$ defined by $\mathbf{v} \rightarrow [\mathbf{v}]_S$, is an isomorphism.

Theorem 3.7 .

$V \cong W \Leftrightarrow \dim(V) = \dim(W)$.

Proof.

$(\Leftarrow)$: $\phi_S : V \rightarrow F^n$ and $\phi'_S : W \rightarrow F^n$ are isomorphism. So $\phi_S^{-1} \circ \phi'_S$ is still an isomorphism. ■

Definition 3.4 .

Let a linear map $f : V \rightarrow W$. Fix B_1 and B_2 as bases of V and W , where $\dim(V) = n$ and $\dim(W) = m$. The matrix is called the matrix representing f w.r.t the bases B_1 and B_2 . We denote it as $[f]_{B_1}^{B_2} = (a_{ij})_{i,j}$. Equivalently, given any vector $\mathbf{v}_j \in V$, we have $f(\mathbf{v}_j) = \sum_{i=1}^m a_{ij} \mathbf{w}_i$.

Draw the fundamental diagram. The bridge between vector space and actual vector world is linear isomorphism.

Theorem 3.8 (Linear maps and matrix).

$$[f]_{B_1}^{B_2} \cdot [\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}$$

Corollary .

Rank and nullity are invariant under different choices of bases. $\mathbf{v} \in \ker(f) \Leftrightarrow f(\mathbf{v}) = \mathbf{0} \Leftrightarrow [f(\mathbf{v})]_{B_2} = [f]_{B_1}^{B_2} \cdot [\mathbf{v}]_{B_1} = \mathbf{0} \Leftrightarrow [\mathbf{v}]_{B_1} \in \ker([f]_{B_1}^{B_2})$. Note that the choice of bases (B_1, B_2) is arbitrary. By dimension formula, we deduce that rank and nullity are preserved.

Theorem 3.9 (Composition and matrix multiplication).

$$[g \circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}$$

Similarly, we can show the connection regarding sum and scaling (linear homomorphism, exercise), inverse function / matrix (later).

Theorem 3.10 (Rank inequality).

Let $f : U \rightarrow V$ and $g : V \rightarrow W$ be two linear maps. Linear map version:

$$\text{rank}(g \circ f) \leq \min\{\text{rank}(f), \text{rank}(g)\}.$$

Matrix version:

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

Theorem 3.11 .

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. If $BA = I_n$ and $AB = I_m$, then $n = m = \text{rank}(A)$.

This theorem shows that only square matrices can possibly be invertible.

Definition 3.5 .

Let A be a n -square matrix. If there exists B such that $AB = BA = I_n$, then we say A is invertible/non-singular. And B is called the inverse (matrix) of A (A^{-1}).

Theorem 3.12 .

$$([f]_{B_1}^{B_2})^{-1} = [f^{-1}]_{B_2}^{B_1}.$$

Check $[id_V]_{B_1}^{B_1} = I_n$. Use the definition of inverse matrix.

3.2 Matrix Operations**Exercise 3.1 .**

Row / column operations:

1. Write down the three types of row operations and define the corresponding elementary row matrix.
2. Write down the three types of column operations and define the corresponding elementary column matrix

Theorem 3.13 .

Let A be a $m \times n$ matrix.

1. Performing a row operation on A is equivalent to left-multiplying an elementary row matrix to A .
2. Performing a column operation on A is equivalent to right-multiplying an elementary column matrix to A .

Theorem 3.14 .

Let A be a $m \times n$ matrix.

1. If Q is an invertible $m \times m$ matrix, then $\text{rank}(QA) = \text{rank}(A)$.
2. If P is an invertible $n \times n$ matrix, then $\text{rank}(AP) = \text{rank}(A)$.

Proof.

Hint: rank inequality. ■

The theorem shows that row / column operations preserves rank.

Exercise 3.2 . 1. Define the row echelon form (REF) and column echelon form (CEF).

2. Prove elementary row matrix and elementary column matrix are invertible (Apply operations reversely).

Theorem 3.15 .

Let A be a $m \times n$ matrix.

1. A can be transformed into a REF by a finite number of row operations. Equivalently, there exists an invertible $m \times m$ matrix Q such that QA is in REF.
2. A can be transformed into a CEF by a finite number of column operations. Equivalently, there exists an invertible $n \times n$ matrix P such that AP is in CEF.

Proof.

An algorithm helps us transform any matrix into echelon form. This is suitable for calculations with concrete examples. Invertible matrix is the product of elementary matrices. ■

Theorem 3.16 .

Let A be a matrix with m columns $\{\mathbf{c}_1, \mathbf{c}_2 \cdots \mathbf{c}_m\}$. Then

1. $\text{im}(A) = \text{span}(\{\mathbf{c}_1, \mathbf{c}_2 \cdots \mathbf{c}_m\})$.
2. $\text{rank}(A)$ equals the maximal number of linearly independent columns of A .
3. $\text{rank}(A)$ equals the number of non-zero columns when A is transformed into a CEF.

Proof.

1. Matrix vector multiplication.

2. By sifting method, the sifted subset forms a basis of $\text{im}(A)$.
3. Sifting method is equivalent to transform A into CEF.

■

Definition 3.6 (Smith Normal Form).

A $m \times n$ matrix is said to be in its Smith normal form if the matrix is both in its REF and CEF. Additionally, all non-zero entries of the matrix are 1.

Theorem 3.17 (QAP Theorem).

Let A be a $m \times n$ matrix. There exists invertible matrices $Q \in M_m(F)$ and $P \in M_n(F)$ s.t. QAP is in its Smith normal form.

Theorem 3.18 .

Let A be a $m \times n$ matrix. Then $\text{rank}(A) = \text{rank}(A^\top)$.

Proof.

$$\text{rank}(A) = \text{rank}(QAP) = \text{rank}(I_r) = r = \text{rank}(P^\top A^\top Q^\top) = \text{rank}(A^\top). \quad \blacksquare$$

Theorem 3.19 .

TFAE:

1. the rank of A ($\dim(\text{im}(A))$).
2. the maximal number of linearly independent columns of A .
3. the number of non-zero columns when A is transformed into a CEF.
4. the rank of $A^\top$.
5. the maximal number of linearly independent rows of A .
6. the number of non-zero rows when A is transformed into a REF.

Proof.

$$\begin{aligned} \text{rank}(A) &= \text{rank}(A^\top) \\ &= \text{maximal \# of linearly independent columns of } A^\top \\ &= \text{maximal \# of linearly independent rows of } A \\ &= \text{\# of non-zero columns in CEF of } A^\top = \text{\# of non-zero rows in REF of } A. \quad \blacksquare \end{aligned}$$

Theorem 3.20 .

Every invertible matrix is a product of elementary matrices.

With the above theorem, we can easily find the inverse of an invertible matrix A .

Theorem 3.21 .

Let A be a n -square matrix. If we conduct row operations to transform the augmented matrix:

$$(A \mid I_n) \xrightarrow{\text{row operation}} (I_n \mid B)$$

Then $B = A^{-1}$.

Proof.

Left multiplying the inverse matrix of A is equivalent to Perform row operations since each invertible matrix is a product of elementary matrices. ■

Exercise 3.3 .

Write down the definition of homogeneous and inhomogeneous linear system of equations. Record the coefficient and augmented matrix.

Theorem 3.22 .

The solution of a homogenous system $(A \mid \mathbf{0})$ coincides $\ker(A)$. Particularly, $\dim(\ker(A)) = (\# \text{ of columns (domain)}) - \text{rank}(A)$.

Theorem 3.23 .

Let $A \in M_{m \times n}(F)$. If $m < n$, then the homogenous system $(A \mid \mathbf{0})$ has a non-trivial / non-zero solution.

Proof.

Intuitively, there are more variables than equations.

$\text{im}(A) \subseteq F^m \rightarrow \text{rank}(A) < m$. By rank-nullity theorem, $\dim(\ker(A)) = n - \text{rank}(A) \geq n - m \geq 1 (\because n > m)$. ■

Theorem 3.24 .

Let $(A \mid \mathbf{b})$ be a general system of equations. The system is consistent (has at least one solution) iff $\text{rank}(A) = \text{rank}(A \mid \mathbf{b})$.

Theorem 3.25 (Principle of linearity).

Suppose $\mathbf{v}$ be a solution to the system of equation $(A \mid \mathbf{b})$, then the general solution of $(A \mid \mathbf{b})$ is

$$\mathbf{v} + \ker(A) = \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \ker(A)\}.$$

Corollary .

The system $(A \mid \mathbf{b})$ has a unique solution iff

1. $\text{rank}(A) = \text{rank}(A \mid \mathbf{b})$.
2. $\ker(A) = \{\mathbf{0}\}$.

Theorem 3.26 (TFAE).

Let $T : V \rightarrow W$ be a linear transformation such that $\dim(V) = \dim(W)$. Having fixed bases, T can be represented by a square matrix A . TFAE:

1. T is injective.
2. $\ker(T) = \{\mathbf{0}\}$.
3. $\text{nullity}(T) = 0$.
4. $\text{rank}(T) = \dim(V) = \dim(W)$.
5. T is surjective.
6. $\text{im}(T) = W$.
7. T is an isomorphism.
8. A is invertible (A^{-1} exists).
9. $\ker(A) = \{\mathbf{0}\} \rightarrow \text{nullity}(A) = 0$.
10. A is full rank ($\text{rank}(A) = \text{rank}(T)$).
11. All of rows of A are linearly independent.
12. All of columns of A are linearly independent.
13. The system of equations $A\mathbf{x} = \mathbf{v}$ has a unique solution, where $\mathbf{x} = A^{-1}\mathbf{v}$
($\because \text{rank}(A) = \text{rank}(A \mid \mathbf{b})$ and $\ker(A) = \{\mathbf{0}\}$).
14. $\det(A) = \det(A^\top) \neq 0$.

3.3 Change of coordinate & Linear Operator

Definition 3.7 (Change of coordinate matrix).

A matrix representing $\text{id}_V : V \rightarrow V$ w.r.t two bases of V . Then the matrix from S_1 basis to S_2 basis is denoted by $[\text{id}_V]_{S_1}^{S_2}$.

Definition 3.8 .

A linear transformation $T : V \rightarrow V$ is called a linear operator on V or an endomorphism on V .

Theorem 3.27 .

Let $T : V \rightarrow V$ be a linear operator and $Q = [id_V]_{S_1}^{S_2}$ be a change of coordinate matrix. Then:

$$[T]_{S_1}^{S_1} = Q^{-1} \cdot [T]_{S_2}^{S_2} \cdot Q.$$

Proof.

Draw the diagram to derive the equation. ■

Application: Diagonalization through $B = Q^{-1}AQ$. That is, choose a good basis (Q) appropriately s.t. B is as simple as possible (ideally a diagonal matrix).

Theorem 3.28 .

Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space V . Suppose S and S' are two bases of V . Then:

$$\det([T]_S^S) = \det([T]_{S'}^{S'}).$$

Definition 3.9 .

Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space V . Define the determinant of T by $\det(T) := \det([T]_S^S)$ for any choice of a basis S on V .

4 Determinants

Exercise 4.1 .

Let $A = (a_{ij})_{i,j} \in M_n(F)$.

1. Write down the definition of determinants $\det(A)$ by Laplace expansion (Consider $n = 1$ and $n \geq 2$).
2. Write down the definition of (i, j) -minor and (i, j) -cofactor.

Remark .

Let $\det : M_n(F) \rightarrow F$.

1. $\det$ is not linear.
2. $\det$ is a multi-linear function.

Example 4.1 .

Let A be an upper triangular matrix. Then $\det(A) = \prod_{i=1}^n a_{ii}$.

Proof.

Prove by induction.

1. When $n = 1$, $\det(A_1) = 1$.
2. Suppose $n = k$ holds. Then for $n = k + 1$, $\det(A_{k+1}) = a_{11}M_{11}$. Note that $M_{1,j} = 0 \ \forall j \neq 1$.
3. So for a matrix A_k , we extend a larger matrix A_{k+1} by the left row and upper column.

■

Theorem 4.1 (Determinants of elementary matrices).

For elementary matrices, we have:

- $\det(E_{p,q}^I) = -1$
- $\det(E_{\lambda,p}^{II}) = \lambda$
- $\det(E_{\mu,p,q}^{III}) = 1$
- $\det(E) = \det(E^\top) \neq 0$

Theorem 4.2 .

Let $A \in M_n(F)$:

1. Exchangeing two rows of A multiplies $\det(A)$ by -1 .
2. Multiplying a row of A by a non-zero $\lambda \in F$ multiplies $\det(A)$ by λ .
3. Adding a multiple of a row to another row does not change $\det(A)$.

Proof.

This proof is very hard.

- For (1), we first prove swapping two rows and generalize to swap any two rows. Express the determinant as a linear combination of some $N_{p,q}$, which are minors deleting 1,2 rows and p,q columns.
- For (2) and (3), it suffices to show $\det : M_n(F) \rightarrow F$ is row-wise linear, or a multi-linear function. That is, determinants is linear on each row with other rows fixed.

■

Theorem 4.3 .

Let $A \in M_n(F)$.

1. If $\det(A)$ contains a zero row, then $\det(A) = 0$.
2. Let $\lambda \in F$. Then $\det(\lambda A) = \lambda^n \det(A)$.

Exercise 4.2 .

Compute $\det(-A)$.

Theorem 4.4 .

A is not invertible if and only if $\det(A) = 0$.

Proof.

$(\Rightarrow)$

1. By theorem (3.26), A is not invertible $\Leftrightarrow \text{REF}(A)$ contains (at least) one zero row $\Leftrightarrow \det(\text{REF}(A)) = 0$.
2. Note $\det(\text{REF}(A)) = \det(E_1 E_2 \cdots E_k A) = \det(E_1) \det(E_2) \cdots \det(E_k) \det(A)$, where determinants of elementary matrices are non-zero. So $\det(A) = 0$.

($\Leftarrow$)

1. Suppose by contradiction, then $\det(A)\det(A^{-1}) = \det(I)$.
2. Logically equivalent to show that A is invertible then $\det(A) \neq 0$. Since every invertible matrix is a product of elementary matrices, whose determinant are non-zeros.

■

Remark .

The above theorem is logically equivalent to the (4.6).

Theorem 4.5 .

Let $A, B \in M_n(F)$, $\det(AB) = \det(A)\det(B)$.

Proof.

If A is invertible, then it is an easy proof. If A is not invertible, we have $\det(A) = 0$. Since $\det(AB) = \det(A)\det(B) = 0$. Since $\text{rank}(AB) \leq \text{rank}(A) < n$, we have $\det(AB) = 0$.

■

Theorem 4.6 .

Let $A \in M_n(F)$.

1. A is invertible $\iff \det(A) \neq 0$.
2. If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$

Theorem 4.7 .

$\det(A^\top) = \det(A)$.

Proof.

If A is invertible, then we can prove through elementary matrices. If A is not invertible, then we compute $\text{rank}(A^\top)$ by QAP and TFAE.

■

This implies column operations on determinants is workable.

Exercise 4.3 .

Write down the general form of Laplace expansion and prove it.

Definition 4.1 (Adjugate Matrix).

Let $A = (a_{ij} \in M_n(F))$. The adjugate matrix of A is defined by $\text{adj}(A) := ((-1)^{i+j}M_{ij})^\top$.

Theorem 4.8 .

Let $A \in M_n(F)$. Then we have:

1. $A \cdot \text{adj}(A) = \det(A) \cdot I_n$.
2. If A is invertible, then $A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$.

Proof.

Refer to the handwritten notes. ■

Theorem 4.9 (Cramer's rule).

Let $A \in M_n(F)$ be an invertible matrix. Then the system of equations $A\mathbf{x} = \mathbf{b}$ has a unique solution (TFAE). Moreover, if $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$, then $x_i = \frac{\det(A_i)}{\det(A)}$, where A_i is the matrix obtained by replace the i -th column of A by the column vector $\mathbf{b}$.

Proof.

$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)}\text{adj}(A)\mathbf{b}$. Spell the expression out. ■

Block Matrix**Exercise 4.4 .**

Write down the definition of 2×2 block matrix. Show the multiplication of two block matrices.

Theorem 4.10 (Block inversion formula).

Suppose:

1. A and D are invertible square matrices.
2. A and D are possibly different sizes.

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{D}^{-1} \end{pmatrix}.$$

In particular, by determinant formula, $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D}) \neq 0$.

Theorem 4.11 (Block determinant formula).

Suppose:

1. A and D are square matrices.
2. A and D are possibly different sizes.

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D}).$$

Proof.

Both formulas need some observations about multiplication of matrices. For the second one, we wish to transform

$$\begin{pmatrix} \mathbf{I} & \mathbf{B} \\ \mathbf{O} & \mathbf{I} \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}.$$

Then apply the following formula. ■

Theorem 4.12 .

Let n be a positive integer and A be a square matrix. Then:

$$\det \begin{pmatrix} \mathbf{I}_n & \mathbf{O} \\ \mathbf{O} & \mathbf{A} \end{pmatrix} = \det \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} = \det(A).$$

Proof.

Prove by induction. ■

Exercise 4.5 .

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size, prove

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} = \det(\mathbf{A} + \mathbf{B})\det(\mathbf{A} - \mathbf{B}).$$

Theorem 4.13 (Schur's factorization).

Suppose

1. $\mathbf{A}$ and $\mathbf{D}$ are square matrices.
2. $\mathbf{A}$ and $\mathbf{D}$ are possibly different sizes.
3. $\mathbf{A}$ is invertible.

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}).$$

Proof.

Observation: To kill $\mathbf{C}$, do row operations with a block of rows at the same time. Consider left multiplying a "block" elementary matrix. ■

5 Diagonalization

5.1 Similarity

Definition 5.1 .

We say A and B of $M_n(F)$ are similar if there exists an invertible $Q \in M_n(F)$ s.t. $B = Q^{-1}AQ$.

Theorem 5.1 (Sufficient condition for similarity).

If $A \sim B$ and $A, B \in M_n(\mathbb{F})$, then

1. $\text{rank}(A) = \text{rank}(B)$.
2. $\text{nullity}(A) = \text{nullity}(B)$.
3. $\det(A) = \det(B)$.
4. $\text{tr}(A) = \text{tr}(B)$.
5. $p_A(x) = p_B(x)$: the same characteristic polynomial.
6. $m_A(x) = m_B(x)$: the same minimal polynomial.

If we further assume that F is algebraically closed (e.g. $\mathbb{C}$), then they have the same:

1. Multi-set of eigenvalues (counted with algebraic multiplicities).
2. algebraic multiplicity of each eigenvalue.
3. geometric multiplicity of each eigenvalue.

Proof.

1. By rank inequality, multiplying an invertible matrix preserves the rank of the original matrix.
2. By rank-nullity theorem.
3. Easy proof.
4. First prove $\text{tr}(AB) = \text{tr}(BA)$, where $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$.
5. Derive $\det(A - xI) = \det(B - xI)$.
6. First prove the polynomial set satisfied by A and the one satisfied by B are equal. Then the monic polynomial of minimal degree satisfied

by the corresponding matrix (i.e. minimal polynomial) is the same.

If we assume $A, B \in M_n(\mathbb{C})$, by the fundamental theorem of algebra,

1. Same characteristic polynomial implies same set of eigenvalues.
2. Similar argument for algebraic multiplicities of eigenvalues.
3. Show $\text{nullity}(A - \lambda I_n) = \text{nullity}(B - \lambda I_n)$.

■

Remark . 1. Multi-set of eigenvalues, which means the repetition of an eigenvalue matters. We will introduce the concept of spectrum later, and the statement is equivalent to say two similar matrix have the same spectrum.

2. $A \sim B \Leftrightarrow A$ and B represent the same linear operator under different bases.

Definition 5.2 (Diagonalizable operators and matrices). 1. A linear operator $T : V \rightarrow V$ is diagonalizable if $\exists$ a basis S of V s.t. the matrix $[T]_S^S$ is diagonal.

2. A matrix $A \in M_n(F)$ is diagonalizable if A is similar to a diagonal matrix, i.e., $\exists$ an invertible matrix Q s.t. $Q^{-1}AQ$ is diagonal.

Definition 5.3 .

Let $T : V \rightarrow V$ be a linear operator.

1. $\lambda \in F$ is an eigenvalue of T if $T(\mathbf{v}) = \lambda \mathbf{v}$ for some non-zero $\mathbf{v} \in V$.
2. $E_\lambda(T) = \ker(A - \lambda \cdot \text{id})$ is called the eigenspace w.r.t the eigenvalue λ .
3. Any **non-zero** $\mathbf{v} \in E_\lambda(T)$ is called an eigenvector of T w.r.t the eigenvalue λ . So any eigenvector satisfies $T(\mathbf{v}) = \lambda \mathbf{v}$.

Theorem 5.2 .

If $\lambda_1, \lambda_2, \dots, \lambda_n$ be distinct eigenvalues of T and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be the corresponding eigenvectors, then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent.

Proof.

Prove by induction. Suppose it holds for $n = k - 1$. Then consider $n = k$. Suppose $a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = 0$ (Equation 1). Apply T on both sides and use the identity $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i \forall i = 1, 2, \dots, k$ (Equation 2). Note that $\lambda_i \neq \lambda_j \forall i \neq j$. Reduce the $k - \text{th}$ term to obtain a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$. By induction hypothesis and distinctness of eigenvalues,

we have $a_1 = a_2 = \cdots a_{k-1} = 0$. Simplify the equation 1 to get $a_k = 0$. By the principle of mathematical induction, the proof is completed. ■

Definition 5.4 (Characteristic polynomial).

Two cases:

1. Let $A \in M_n(F)$. The characteristic polynomial of A is defined by $p_A(x) = \det(A - xI_n)$.
2. Let $T : V \rightarrow V$ be a linear operator. The characteristic polynomial of T is defined by $p_T(x) = \det([T]_s^s - xI_n)$ (For any basis s of V).

Theorem 5.3 .

Eigenvalues are the roots of the characteristic polynomial of A . That is, λ is an eigenvalue of A iff $p_A(\lambda) = \det(A - \lambda I_n) = 0$.

Remark .

The above theorem demonstrates how to find eigenvalues practically.

5.2 Diagonalizability

Theorem 5.4 (Diagonalizability Criterion I: Eigen-basis).

$A \in M_n(F)$ is diagonalizable iff $\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n$ in F^n s.t. each v_i is an eigenvector of A w.r.t. some eigenvalue λ_i .

Proof.

This is the definition of diagonalizability itself. Associate A with a linear operator $f_A : F^n \rightarrow F^n$. Let B be the standard basis.

1. Think why $A = [f_A]_B^B$.
2. f_A is diagonalizable iff $\exists$ a basis S of F^n s.t. $[f_A]_S^S$ is diagonal. $\Leftrightarrow A$ is diagonalizable iff $\exists$ an invertible $Q = [id_V]_S^B$.
3. Think: Q is essentially formed by putting each eigenvectors in S as columns.
4. Think: Why S is a set of eigenvectors? Why the diagonal entries in $[f_A]_S^S$ are the corresponding eigenvalues? ($f_A(\mathbf{v}_i) = A\mathbf{v}_i = \lambda\mathbf{v}_i$).

■

Theorem 5.5 .

If $A \in M_n(F)$ has n distinct eigenvalues, then A is diagonalizable.

Proof.

1. Distinct eigenvalues shows that ...?
2. By replacement theorem, we find exactly a set of eigen-basis, hence A is diagonalizable.

■

Definition 5.5 .

Let $A \in M_n(\mathbb{C})$. By the fundamental theorem of algebra, $p_A(x)$ can be factorized as $(-1)^n(x - \lambda_1)^{r_1}(x - \lambda_2)^{r_2} \cdots (x - \lambda_k)^{r_k}$, where $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct. For each eigenvalue λ_i ,

1. Algebraic multiplicity of λ_i : # of copies of $(x - \lambda_i)$ in $p_A(x)$, denoted by $a(\lambda_i) = r_i$.
2. Geometric multiplicity of λ_i , denoted by $g(\lambda_i) = \text{nullity}(A - \lambda_i I_n)$, which measures the dimension of the eigenspace of λ_i .

Theorem 5.6 .

Let $A \in M_n(\mathbb{C})$. For each λ of A :

$$1 \leq g(\lambda) \leq a(\lambda).$$

Proof.

1. $1 \leq g(\lambda)$: $\det(A - \lambda I_n) = 0$ and use TFAE.
2. $g(\lambda) \leq a(\lambda)$: Take a basis of $(A - \lambda I_n)$ (assume dimension equals p). By basis extension theorem, we can obtain a basis S of $\mathbb{C}^n$. Apply associated linear operator T and get a "quasi-diagonal" matrix. Note that $p_{[T]_S^S}(x) = (-1)^n(x - \lambda)^p \times r(x)$. $r(x)$ may or may not contain $(x - \lambda)$.

■

Theorem 5.7 (Diagonalizability Criterion II: Multiplicity).

Let $A \in M_n(\mathbb{C})$. TFAE:

1. A is diagonalizable.
2. $\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2 \dots \mathbf{v}_n$ in $\mathbb{C}^n$ s.t. each v_i is an eigenvector of A w.r.t. some eigenvalue λ_i .
3. For each λ of A , we have $g(\lambda) = a(\lambda)$.

Proof.

It suffices to show:

1. (1) implies (3): Let D be a diagonal matrix, $A \sim D$ implies they have the same geometric multiplicities.
2. (3) implies (2): For each eigenspace, choose S_i to be a basis of $E_{\lambda_i}(A)$. Claim that $S = S_1 \cup S_2 \cup \dots \cup S_k$ forms an eigen basis of A . Detail: for each λ_i , we may choose more than one eigenvector, why is the whole set S linearly independent?

■

Exercise 5.1 .

Take a polynomial $f \in F[x]$. For $A \in M_n(F)$, define $f(A)$.

Theorem 5.8 (Cayley-Hamilton Theorem).

Let $A \in M_n(F)$ and its characteristic polynomial $p_A(x)$. Then $p_A(A) = \mathbf{O}$ (the zero matrix).

Proof.

Use $(A - xI)\text{adj}(A - xI) = \det(A - xI)I = p_A(x)I$. Compare coefficients.

■

Definition 5.6 (Minimal Polynomial).

Let $A \in M_n(\mathbb{C})$. The minimal polynomial is the monic polynomial of minimal degree s.t. $m_A(A) = \mathbf{O}$.

Monic means the leading coefficient is 1.

Theorem 5.9 .

Let $A \in M_n(\mathbb{C})$, $p_A(x)$ and $m_A(x)$ be its characteristic polynomial and minimal polynomial respectively.

1. For any $f \in \mathbb{C}[x]$ satisfying $f(A) = \mathbf{O}$, then $f(x)$ is divisible by $m_A(x)$.

2. $p_A(x)$ is divisible by $m_A(x)$ / $m_A(x)$ divides $p_A(x)$.
3. Each eigenvalue λ of A is a root of $m_A(x)$ / $m_A(\lambda) = 0$.

Proof.

1. Conduct division of $f(x)$ by $m_A(x)$. Aim to argue the remainder $r(x)$ must be zero polynomial. Suppose not, then $r(x)$ is a smaller degree polynomial satisfying $r(A) = 0$, which yields a contradiction.
2. By Cayley-Hamilton and the property (1).
3. Use (5.10), we have: $m_A(A)\mathbf{v} = m_A(\lambda)\mathbf{v}$. This forces $m_A(\lambda) = 0$.

■

Theorem 5.10 .

Let f be a non-constant polynomial. If λ is an eigenvalue of A , then $f(\lambda)$ is an eigenvalue of $f(A)$.

Proof.

Key observation: $A^k\mathbf{v} = \lambda^k\mathbf{v}$ for any $k \in \mathbb{N}$.

■

Lemma .

Every matrix $A \in M_n(\mathbb{C})$ has a unique minimal polynomial.

Proof.

Use the property (1).

■

Lemma (Candidate of minial polynomial).

Let $A \in M_n(\mathbb{C})$ and $\lambda_1, \lambda_2, \dots, \lambda_k$ be all distinct eigenvalues of A . If $p_A(x) = (-1)^n \cdot (x - \lambda_1)^{r_1}(x - \lambda_2)^{r_2} \dots (x - \lambda_k)^{r_k}$. Then $m_A(x) = (x - \lambda_1)^{s_1}(x - \lambda_2)^{s_2} \dots (x - \lambda_k)^{s_k}$, where $1 \leq s_i \leq r_i$ for each $i = 1, 2, \dots, k$.

Proof.

Property (2) and (3) imply this lemma.

■

Theorem 5.11 (Diagonalizability Criterion III: Minimal Polynomial).

Let $A \in M_n(\mathbb{C})$. TFAE:

1. A is diagonalizable.

2. $\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in $\mathbb{C}^n$ s.t. each $\mathbf{v}_i$ is an eigenvector of A w.r.t. some eigenvalue λ_i .
3. For each λ of A , we have $g(\lambda) = a(\lambda)$.
4. The minimal polynomial $m_A(x)$ has no repeated factor.

Proof.

1. (2) implies (4): Suppose (2) is true. Let $p_A(x) = (-1)^n \cdot (x - \lambda_1)^{r_1} (x - \lambda_2)^{r_2} \cdots (x - \lambda_k)^{r_k}$ and $q(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$, where $\lambda_i \neq \lambda_j \forall i \neq j$. Aim to show $q(x) = m_A(x)$. By property (3) in (5.9), we have $q(x) \mid m_A(x)$. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\mathbb{C}^n$ and eigenvectors of A . For each $\mathbf{v} \in S$, $A\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda = \lambda_i$. By (5.10), $q(A)\mathbf{v} = q(\lambda)\mathbf{v}$ and $q(\lambda) = 0$. Note that $S \subseteq \ker(q(A)) \rightarrow |S| \leq \text{nullity}(q(A)) \leftrightarrow \text{rank}(q(A)) = 0 \leftrightarrow q(A) = \mathbf{O}$. So by property (1) in (5.9), $m_A(x) \mid q(A)$.

2. Suppose $m_A(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$, where $\lambda_i \neq \lambda_j \forall i \neq j$. By definition, $m_A(A) = \mathbf{O}$. Use (1), we have:

$$\text{nullity}\left(\prod_{i=1}^k (A - \lambda_i I)\right) \leq \sum_{i=1}^k \text{nullity}(A - \lambda_i I) \leq \sum_{i=1}^k g(\lambda_i) = n.$$

However, $\text{nullity}(m_A(A)) = \text{nullity}(\mathbf{O}) = n$. This forces the equality.

■

Proposition 1 .

Let $B, C \in M_n(F)$, then $\text{nullity}(BC) \leq \text{nullity}(B) + \text{nullity}(C)$.

Proof.

First, show that $\ker(C) \subseteq \ker(BC)$. Extend a basis of $\ker(C)$, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ into a basis of $\ker(BC)$, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p\}$.

Next, show $\{C\mathbf{w}_1, C\mathbf{w}_2, \dots, C\mathbf{w}_p\} \subseteq \ker(B)$. In particular, $\{C\mathbf{w}_1, C\mathbf{w}_2, \dots, C\mathbf{w}_p\}$ is linearly independent. (Show the linear combination lies in $\ker(C)$, and use the linear independence of the basis in $\ker(BC)$).

■

Theorem 5.12 (Schur Triangulation).

Every square matrix $A \in M_n(\mathbb{C})$ is similar to an upper triangular matrix.

Proof.

Prove by induction. Left as an exercise. ■

Definition 5.7 (Spectrum).

The spectrum of a linear operator T is the multi-set contains the all eigenvalues of T , denoted by $\text{spec}(T) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$. Note that these elements might repeat.

Theorem 5.13 (Spectral mapping theorem).

Let $A \in M_n(\mathbb{C})$ and $f \in \mathbb{C}[x]$ be a non-constant polynomial.

1. If λ is an eigenvalue of A , then $f(\lambda)$ is an eigenvalue of $f(A)$.
2. If μ is an eigenvalue of $f(A)$, then $\exists$ an eigenvalue λ of A s.t. $\mu = f(\lambda)$.

That is, if $\text{spec}(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, then $\text{spec}(f(A)) = \{f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)\}$. So spectrum mapping is a bijection between multi-sets.

Proof.

Key result is that diagonal entries of upper triangular matrix is eigenvalues.

Use Schur triangulation to deduce the spectrum mapping. ■

5.3 Application of Diagonalization

Definition 5.8 (T-invariant subspace).

Let $T : V \rightarrow V$ be a linear operator. A subspace $W \subseteq V$ is said to be a T-invariant subspace if $T(W) \subseteq W$. Moreover, the restriction of T on W is defined by $T|_W : W \rightarrow W$, where $T|_W(\mathbf{w}) = T(\mathbf{w}) \forall \mathbf{w} \in W$.

Theorem 5.14 .

Let $W \subseteq V$ be an T-invariant subspace and let $T|_W : W \rightarrow W$ be the restriction. Then $m_{T|_W}(x)$ divides $m_T(x)$.

Proof.

Key observation: $m_T(T) = \mathbf{0}$ implies $m_T(T|_W) = \mathbf{0}$. Use the property (1) in (5.9). ■

Exercise 5.2 .

Verify that

1. $\ker(T|_W) = \ker(T) \cap W$.
2. $\text{im}(T|_W) = \{T(\mathbf{w}) : \mathbf{w} \in W\}$.

Theorem 5.15 (Simultaneous Eigenvector).

Let $T_1, T_2 : V \rightarrow V$ be two commutable linear operators of a *complex* finite-dimensional V , then there exists $\mathbf{v}$ s.t. $\mathbf{v}$ is simultaneously an eigenvector of both T_1 and T_2 .

Proof.

Let $E_\lambda(T_1)$ be an eigenspace of T_1 .

1. Claim that $E_\lambda(T_1)$ is T_2 invariant.
2. Consider the restriction $T_2|_{E_\lambda(T_1)} : E_\lambda(T_1) \rightarrow E_\lambda(T_1)$. Take an eigenvector $\mathbf{v}$ of $T_2|_{E_\lambda(T_1)}$ w.r.t. some eigenvalue λ' . This vector is simultaneously an eigenvector of $E_\lambda(T_1)$.

■

Theorem 5.16 (Simultaneous Diagonalizability).

Let $T_1, T_2 : V \rightarrow V$ be two commutable and diagonalizable linear operators of a *complex* finite-dimensional V , then there exists a simultaneous eigen-basis S s.t. $[T_1]_S^S$ and $[T_2]_S^S$ are both diagonal and hence simultaneously diagonalizable.

Proof.

This is just a stronger version of the previous theorem.

■

Definition 5.9 (Exponential of Matrix).

Let $A \in M_n(\mathbb{C})$. The exponential of A is defined by:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Theorem 5.17 .

If $D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$ is a diagonal matrix, then $e^D = \begin{pmatrix} e^{d_1} & 0 & \cdots & 0 \\ 0 & e^{d_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{d_n} \end{pmatrix}$

Theorem 5.18 .

Let $A, B \in M_n(\mathbb{C})$, then:

1. e^A converges absolutely.
2. If $AB = BA$, then $e^{A+B} = e^A e^B$.

Proof.

The above theorems are stated without proof, as they may require knowledge of normed spaces, which is beyond the current scope. ■

Theorem 5.19 .

Let $A \in M_n(\mathbb{C})$.

1. If $A \sim B$, then $e^A \sim e^B$.
2. Consider a function $f : \mathbb{C} \rightarrow M_n(\mathbb{C})$ defined by $f(t) = e^{At}$, then

$$f'(t) = \frac{d}{dt}e^{At} = Ae^{At}.$$

Corollary .

Give a system of differential equations:

$$\frac{d}{dt}\mathbf{v}(t) = A\mathbf{v}(t),$$

where $A \in M_n(\mathbb{C})$, then the solution is given by:

$$\mathbf{v}(t) = e^{At}\mathbf{v}(0).$$

Remark .

To compute e^{At} , we need to address higher powers of A , then diagonalization or Cayley-Hamilton is useful here.

Exercise 5.3 .

Give a system of difference equations:

$$\mathbf{v}_n = A\mathbf{v}_{n-1},$$

where $A \in M_n(\mathbb{C})$, then the solution is given by:

$$\mathbf{v}_n = A^n\mathbf{v}_0.$$

6 Linear Duality

Let V and W be two finite-dimensional vector spaces over a field $\mathcal{F}$ s.t. $\dim(V) = n$ and $\dim(W) = m$. Recall our *dream diagram*:

$$\begin{array}{ccc} V & \xrightarrow{\mathcal{L}(V,W)} & W \\ \phi_S \downarrow & & \downarrow \phi_{S'} \\ \mathcal{F}^n & \xrightarrow{\mathcal{M}_{m \times n}(\mathcal{F})} & \mathcal{F}^m \end{array}$$

Fix S as a basis of V , the coordinate mapping $\phi : V \rightarrow \mathcal{F}^n$ defined by $\mathbf{v} \mapsto [\mathbf{v}]_S$ is an isomorphism. So does the coordinate mapping $\phi_{S'}$.

Now consider the vector space of all linear functions from V to W , denoted by $\mathcal{L}(V, W) = \{f : V \rightarrow W : f \text{ is linear}\}$, then we have:

Theorem 6.1 .

Fix bases S and S' of V and W respectively, then there exists an isomorphism of vector spaces:

$$\Phi_S^{S'} : \mathcal{L}(V, W) \rightarrow \mathcal{M}_{m \times n}(\mathcal{F}),$$

defined by $f \mapsto [f]_S^{S'}$.

Corollary .

If both V, W are finite dimensional over $\mathcal{F}$, then

$$\dim_{\mathcal{F}}(\mathcal{L}(V, W)) = \dim_{\mathcal{F}}(V)\dim_{\mathcal{F}}(W).$$

Next, we apply the above result to the special case when $W = \mathcal{F}$.

Definition 6.1 (Dual Space).

Let V be a vector space over a field $\mathcal{F}$. The dual space of V , denoted by V^* , is defined by:

$$V^* = \mathcal{L}(V, \mathcal{F}) = \{f : V \rightarrow \mathcal{F} : f \text{ is linear}\}.$$

Take any element $f \in V^*$, we call $f : V \rightarrow \mathcal{F}$ is linear functional on V .

Theorem 6.2 .

Fix a basis S of V and $B = \{1\}$ be a standard basis of $\mathcal{F}$, then there exists an isomorphism of vector spaces:

$$\Phi_S^B : \mathcal{L}(V, \mathcal{F}) \rightarrow \mathcal{M}_{1 \times n}(\mathcal{F}),$$

defined by $f \mapsto [f]_S^B$.

Corollary .

If both V is finite dimensional over $\mathcal{F}$, then

$$\dim_{\mathcal{F}}(V^*) = \dim_{\mathcal{F}}(V).$$

Definition 6.2 (Dual Basis).

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of V . $S^* = \{\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_n^*\}$ is said to be the dual basis of S in V^* if $\phi_S^B(\mathbf{v}_k^*) = e_k$ for each $k = 1, 2, \dots, n$.

Lemma .

Fix any basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of V , then its dual basis $S^* = \{\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_n^*\}$ satisfies, for each i :

$$\begin{cases} \mathbf{v}_i^*(\mathbf{v}_j) = 1, & \text{if } i = j, \\ \mathbf{v}_i^*(\mathbf{v}_j) = 0, & \text{if } i \neq j. \end{cases}$$

Proof.

For $i = 1$, consider $\mathbf{v}_1^* : V \rightarrow \mathcal{F}$. By definition, $\Phi_S^B(\mathbf{v}_1^*) = e_1$, i.e., $[\mathbf{v}_1^*]_S^B = (1, 0, \dots, 0)$. This implies that $\mathbf{v}_1^*(\mathbf{v}_1) = 1 \cdot 1$ and $\mathbf{v}_1^*(\mathbf{v}_j) = 0 \cdot 1$ for $j \neq 1$. Now repeat the above argument for $i = 2, 3, \dots, n$. ■

Definition 6.3 (Dual linear maps).

Let $T : V \rightarrow W$ be a linear map btw two vector spaces V and W . The dual of T is a linear map btw the dual spaces of V and W :

$$T^* : W^* \rightarrow V^* \text{ given by } w^* \mapsto w^* \circ T.$$

Theorem 6.3 .

Let $T : V \rightarrow W$ be a linear map btw two finite-dimensional vector spaces V and W . Fix bases S and S' of V and W respectively, then

$$[T^*]_{S'^*}^{S^*} = ([T]_S^{S'})^T.$$

Proof.

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of V . Let $S' = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be a basis of

W . Suppose $[T]_S^{S'} = A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$, we want to show $[T^*]_{S'^*}^{S^*} =$

$A^\top$.

Let $\mathbf{v} = \sum_{k=1}^n c_k \mathbf{v}_k$. Consider $T^*(\mathbf{w}_1^*)(\mathbf{v}) = \mathbf{w}_1^* \circ T(\mathbf{v}) =$

$\mathbf{w}_1^*(\sum_{k=1}^n c_k T(\mathbf{v}_k)) = \mathbf{w}_1^*(\sum_{k=1}^n c_k \sum_{j=1}^m a_{jk} \mathbf{w}_j) = \sum_{k=1}^n c_k a_{1k} \cdot 1$. Note that $c_k = v_k^*(\mathbf{v})$, so we get $T^*(\mathbf{w}_1^*) = b_1^* \circ T(\mathbf{v}) = \sum_{k=1}^n a_{1k} v_k^*(\mathbf{v})$. ■

Definition 6.4 (Annihilator).

Let V be a vector space and $U \subseteq V$ be a subspace. The annihilator of U is a subspace of V^* defined by:

$$U^0 = \{f \in V^* : f(\mathbf{u}) = 0 \ \forall \mathbf{u} \in U\}.$$

Theorem 6.4 (Linear Duality Theorem (I)).

Let V be a finite dimensional vector space and $U \subseteq V$, then:

$$\dim(V) = \dim(U) + \dim(U^0).$$

Proof.

Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis of U . By basis extension theorem, we can extend it into $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n\}$. So $\{\mathbf{u}_1^*, \mathbf{u}_2^*, \dots, \mathbf{u}_k^*, \mathbf{u}_{k+1}^*, \dots, \mathbf{u}_n^*\}$ is the dual basis in V^* . Next, we want to show $\{\mathbf{u}_{k+1}^*, \dots, \mathbf{u}_n^*\}$ forms a basis of U^0 .

1. For each $k+1 \leq j \leq n$, $\mathbf{u}_j^*(\mathbf{u}_i) = 0$ for $1 \leq i \leq k$. So $\mathbf{u}_j^* \in U^0$.
2. Let $\theta \in U^0 \subseteq V^*$, so $\theta = \sum_{i=1}^n a_i \mathbf{u}_i^*$, and we wish $a_1 = a_2 = \dots = a_k = 0$. Let $\mathbf{u} \in U$, so $\mathbf{u} = \sum_{i=1}^k \mathbf{u}_i$. By definition, $\theta(\mathbf{u}_j) = 0$ forces $\sum_{i=1}^k a_i \mathbf{u}_i^*(\mathbf{u}_j) = 0$ for $1 \leq j \leq k$.
3. Linear independence is trivial.

By the claim, we have $\dim(U^0) = n - k = \dim(V) - \dim(U)$. ■

Theorem 6.5 (Linear Duality Theorem (II)).

Let $T : V \rightarrow W$ be a linear map btw finite dimensional vector spaces. Then we have:

1. $\ker(T^*) = (\text{im}(T))^0$.
2. $\text{rank}(T) = \text{rank}(T^*)$.
3. $\text{im}(T^*) = (\ker(T))^0$.

Proof.

Left as an exercise. ■

Corollary .

(6.5) implies that

1. T^* is injective iff T is surjective.
2. $\text{rank}(A) = \text{rank}(A^\top)$.
3. T^* is surjective iff T is injective.

Proof.

1. $\text{im}(T) = W \leftrightarrow \dim((\text{im}(T))^0) = 0 = \dim(\ker(T^*))$.
2. Recall theorem (6).
3. $\ker(T) = \{\mathbf{0}\} \leftrightarrow \dim((\ker(T))^0) = \dim(\text{im}(T^*)) = \dim(V) = \dim(V^*)$

■

Definition 6.5 (Double Dual).

The dual of a dual space is defined by:

$$(V^*)^* = \{g : V^* \rightarrow F : g \text{ is linear}\}.$$

Theorem 6.6 (Linear Duality Theorem (III)).

Let $ev : V \rightarrow (V^*)^*$ defined by $\mathbf{v} \mapsto ev(\mathbf{v})$. That is, for each $\mathbf{v} \in V$, $ev(\mathbf{v}) : V^* \rightarrow F$ is a linear functional on V^* . So the ev function satisfies that for $f \in V^*$, $ev(\mathbf{v})(f) = f(\mathbf{v})$. Under this setting, we have:

1. ev is an injective linear map.
2. If V is finite dimensional, then ev is an isomorphism btw V and $(V^*)^*$.

Proof.

1. Take $\mathbf{v} \in \ker(ev)$, we have $ev(\mathbf{v}) = \mathbf{0}$. That is, $ev(\mathbf{v})(f) = f(\mathbf{v}) = 0$ for $f \in V^*$. Since f is arbitrary in the dual space V^* , this forces $\mathbf{v} = \mathbf{0}$ and $\ker(ev) = \{\mathbf{0}\}$. This proves the injectiveness.
2. Note that $\dim(V) = \dim(V^*) = \dim((V^*)^*)$, $ev : V \rightarrow (V^*)^*$ is injective linear map from 1, by (3.26), ev is also a surjective map and hence an isomorphism.

■

Remark .

If V is infinite dimensional, then ev might not be surjective in general, which is beyond the current scope.

7 Jordan Canonical Form

Definition 7.1 (Jordan Block).

A Jordan block $J_{\lambda,k}$ of eigenvalue λ and index/size k is a $k \times k$ upper triangular matrix such that $\dots$

Theorem 7.1 .

Let $J = J_{\lambda,k} \in M_k(\mathbb{C})$, then:

1. $p_J(x) = (\lambda - x)^k$.
2. $a_J(\lambda) = k$, and $g_J(\lambda) = 1$.
3. For $1 \leq i \leq k$, $\text{rank}((J - \lambda I_k)^i) = k - i$ and $\text{nullity}((J - \lambda I_k)^i) = i$.
4. $\text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1}) = \begin{cases} 1, & \text{if } 1 \leq i \leq k \\ 0, & \text{if } i \geq k + 1 \end{cases}$.
5. $m_J(x) = (x - \lambda)^k$.

Proof.

1,2 is trivial by inspection. For 3 and 4, let $N_k = J - \lambda I_k$. The key observation is that the diagonal of N_k^i is shifted by one unit to the right after multiplying N_k itself. For 5, since $N_k^i \neq \mathbf{0}$ for $i < k$, the minimal polynomial of J must be the characteristic polynomial of J . ■

Remark .

2 and 5 shows that J is diagonalizable only when J is of size 1.

Definition 7.2 .

Direct sum of matrices Let $A \in M_m(F)$, $B \in M_n(F)$, then the direct sum $A \oplus B$ is a block diagonal matrix in $M_{m+n}(F)$ such that

$$A \oplus B = \begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{pmatrix}$$

Exercise 7.1 .

Write down the definition of a direct sum of two subspaces. They are two different concepts.

Theorem 7.2 .

Let $A \in M_m(F)$, $B \in M_n(F)$, then:

1. $\text{rank}(A \oplus B) = \text{rank}(A) + \text{rank}(B)$
2. $\text{nullity}(A \oplus B) = \text{nullity}(A) + \text{nullity}(B)$
3. $p_{A \oplus B}(x) = p_A(x) \cdot p_B(x)$
4. $a_{A \oplus B}(\lambda) = a_A(\lambda) + a_B(\lambda)$
5. $g_{A \oplus B}(\lambda) = g_A(\lambda) + g_B(\lambda)$
6. $m_{A \oplus B}(x) = \text{lcm}\{m_A(x), m_B(x)\}$

Proof.

1–5 are left as an exercise. For 6, observe that for any polynomial $f(x) \in F[x]$, one has $f(A \oplus B) = f(A) \oplus f(B)$. Let $m_{A \oplus B} = f(x)$, $f(A \oplus B) = \mathbf{0}$ by the definition of minimal polynomial, which forces $f(x)$ is a common multiple of $m_A(x)$ and $m_B(x)$. For the "least" part, we provide two methods. Suppose $g(x)$ is another common multiple, we want to show f divides g . Since $m_A(x) \mid g(x)$ and $m_B(x) \mid g(x)$, $g(A \oplus B) = \mathbf{0}$ implies $f \mid g$.

Alternatively, conduct the division $g(x) = f(x) \cdot q(x) + r(x)$. It suffices to show $r(x) \equiv 0$. Suppose, by contradiction, $\deg(r) < \deg(f)$. However, $r(A) = r(B) = \mathbf{0}$ violates the minimality. ■

Definition 7.3 (JCF).

A matrix J is in its Jordan Canonical Form if it is a direct sum of Jordan blocks:

$$J = J_{\lambda_1, k_1} \oplus J_{\lambda_2, k_2} \oplus \cdots \oplus J_{\lambda_n, k_n}.$$

Corollary . 1. Any diagonal matrix is exactly in its JCF. . .

2. Write down the $p_J(x)$ and $m_J(x)$ in terms of $p_{J_{\lambda_i, k_i}}(x)$ and $m_{J_{\lambda_i, k_i}}(x)$.

Theorem 7.3 .

Let J in its JCF and fix $\lambda \in \text{Spec}(J)$, then:

1. # of Jordan blocks of eigenvalue $\lambda = g_J(\lambda)$.
2. # of Jordan blocks of eigenvalue λ of index(size) at least $i = \text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1})$.

Proof.

Let $J = J_{\lambda_1, k_1} \oplus J_{\lambda_2, k_2} \oplus \cdots \oplus J_{\lambda_n, k_n}$. Suppose $\lambda_1 = \lambda_2 = \cdots = \lambda_l = \lambda$, $l \leq n$, $\lambda \notin \{\lambda_{l+1}, \lambda_{l+2}, \dots, \lambda_n\}$. Let $N_k = J - \lambda I_k$, recall (3) and (4) in theorem (7.1),

$$1. g_J(\lambda) = \sum_{i=1}^n \text{nullity}(N_{\lambda_i, k_i}) = \underbrace{1 + 1 + \cdots + 1}_{l \text{ times}} + \underbrace{0 + 0 + \cdots + 0}_{\lambda \neq \lambda_i, i \geq l} = l,$$

which equals # of Jordan blocks of eigenvalue λ .

2. Because $N_k = N_{k_1} \oplus N_{k_2} \oplus \cdots \oplus N_{k_l} \oplus T$, where T is some direct sum of Jordan blocks of eigenvalue that is not λ . Fix i , $\text{nullity}(N_k^i) - \text{nullity}(N_k^{i-1}) = \sum_{j=1}^l (\text{nullity}(N_{k_j}^i) - \text{nullity}(N_{k_j}^{i-1})) = \#\{1 \leq j \leq l : i \leq k_j\}$, which equals # of Jordan blocks of eigenvalue λ of index(size) at least i .

■

Combine the definition of JCF and the theorem (7), we can define the JCF of a general matrix $A \in M_n(\mathbb{C})$.

Definition 7.4 (JCF of a general matrix).

Let $A \in M_n(\mathbb{C})$, the JCF of A is a matrix of J_A s.t.

1. J_A is a direct sum of Jordan blocks.
2. # of Jordan blocks of eigenvalue $\lambda = g_A(\lambda)$.
3. # of Jordan blocks of eigenvalue λ of index(size) at least $i = \text{nullity}((A - \lambda I_k)^i) - \text{nullity}((A - \lambda I_k)^{i-1})$.

Remark .

How to find the JCF of A ?

1. Find spectrum of A via $p_A(x)$.
2. Find $g_A(\lambda)$ to determine # of Jordan blocks.
3. Find $r_i = \text{nullity}((A - \lambda I_k)^i) - \text{nullity}((A - \lambda I_k)^{i-1})$ to determine individual size of a Jordan block.

Let $T : V \rightarrow V$ be a linear operator over a finite-dimensional complex vector space V . In the context of diagonalization, we aim to find an eigenbasis η such that $[T]_\eta^\eta$ is a diagonal matrix. Likewise, our goal now is to find a basis γ such that

$[T]_\gamma^\gamma$ is in its JCF. We first look at a part of matrix corresponding to an eigenvalue λ . Suppose there exists such a part of basis $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, and we put each elements by columns to form a matrix P , then we have:

$$[T]_\beta^\beta P = P [T]_\beta^\beta$$

$$T([\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k]) = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k] \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix},$$

where B is the standard basis. Comparing the columns on both sides yields the following system of equations:

$$\begin{array}{ll} T(\mathbf{v}_1) = \lambda \mathbf{v}_1 & (T - \lambda \text{id})(\mathbf{v}_1) = \mathbf{0} \\ T(\mathbf{v}_2) = \mathbf{v}_1 + \lambda \mathbf{v}_2 & (T - \lambda \text{id})(\mathbf{v}_2) = \mathbf{v}_1 \\ T(\mathbf{v}_3) = \mathbf{v}_2 + \lambda \mathbf{v}_3 & \iff (T - \lambda \text{id})(\mathbf{v}_3) = \mathbf{v}_2 \\ \vdots & \vdots \\ T(\mathbf{v}_k) = \mathbf{v}_{k-1} + \lambda \mathbf{v}_k & (T - \lambda \text{id})(\mathbf{v}_k) = \mathbf{v}_{k-1} \end{array}$$

Observe that:

1. $\mathbf{v}_1$ is the only eigenvector among β .
2. Given $\mathbf{v}_k$, we can generate a chain of β by setting $\{\mathbf{v}_k, (T - \lambda \text{id})(\mathbf{v}_k), (T - \lambda \text{id})(\mathbf{v}_{k-1}), \dots, (T - \lambda \text{id})(\mathbf{v}_2)\}$, or equivalently, $\{\mathbf{v}_k, (T - \lambda \text{id})(\mathbf{v}_k), (T - \lambda \text{id})^2(\mathbf{v}_k), \dots, (T - \lambda \text{id})^{k-1}(\mathbf{v}_k)\}$.
3. $(T - \lambda \text{id})^k(\mathbf{v}_k) = (T - \lambda \text{id})(\mathbf{v}_1) = \mathbf{0}$, so $\mathbf{v}_k \in \ker(T - \lambda \text{id})^k \setminus \ker(T - \lambda \text{id})^{k-1} (\because \mathbf{v}_1 \neq \mathbf{0})$.

These motivates the definition of Jordan chain:

Definition 7.5 (Jordan Chain).

Let $T : V \rightarrow V$ be a linear operator and $\lambda \in \text{Spec}(T)$. We say a set of non-zero vectors $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a Jordan chain of T corresponding to the eigenvalue λ of length k if $\dots$, where $\mathbf{v}_k$ is the end vector and $\mathbf{v}_1$ is the initial vector.

Definition 7.6 (Generalized eigenspace).

Let $T : V \rightarrow V$ be a linear operator and λ be an eigenvalue of T . Define

$$K_\lambda \equiv \bigcup_{i \geq 1} \ker(T - \lambda)^i$$

to be the generalized eigenspace of T w.r.t eigenvalue λ . Any non-zero vector in K_λ is called a generalized eigenvector.

Theorem 7.4 .

Suppose m is the smallest positive integer such that $\ker(T - \lambda I)^m = \ker(T - \lambda I)^{m+1}$. Then

1. we have a strict inclusions of subspaces:

$$\{\mathbf{0}\} \subset \ker(T - \lambda I) \subset \ker(T - \lambda I)^2 \subset \cdots \subset \ker(T - \lambda I)^m.$$

2. $K_\lambda = \ker(T - \lambda I)^m$.

3. Let $p \leq m$ and $\mathbf{v} \in \ker(T - \lambda I)^p \setminus \ker(T - \lambda I)^{p-1}$.

The set $\{\mathbf{v}, (T - \lambda I)(\mathbf{v}), \cdots, (T - \lambda I)^{p-1}(\mathbf{v})\}$ forms a Jordan chain of T w.r.t eigenvalue λ of length p .

Theorem 7.5 .

Let λ be an eigenvalue of T . If $\beta_1, \beta_2, \cdots, \beta_q$ are Jordan chains of λ such that their initial vectors are linearly independent, then $\gamma_\lambda = \beta_1 \cup \beta_2 \cup \cdots \cup \beta_q$ is linearly independent.

Proof.

Let $|\gamma_\lambda| = n$, we aim to do induction on the size of union of Jordan chains, which are linearly independent. If $n = 1$, there are nothing to prove. Suppose $n < k$ is true, then for $n = k$, let $W = \text{span}(\gamma_\lambda)$, we want to show $\dim(W) = k$. Consider a restriction map $T - \lambda I|_W : W \rightarrow W$ and a set $\gamma'_\lambda = \beta'_1 \cup \beta'_2 \cup \cdots \cup \beta'_q$ obtained by removing the end vector of each Jordan chain. By induction hypothesis, γ'_λ is linearly independent. Note that γ_λ spans W , $(T - \lambda I)(\gamma_\lambda) = \gamma'_\lambda$ spans $(T - \lambda I)(W)$, i.e. $\text{im}(T - \lambda I|_W)$. So γ'_λ forms a basis of $\text{im}(T - \lambda I|_W)$ and $\text{rank}(T - \lambda I|_W) = k - q$. Hence

$$\underbrace{k}_{|\gamma_\lambda|} \geq \dim(W) = \text{rank}(T - \lambda I|_W) + \text{nullity}(T - \lambda I|_W) \geq k - q + q = k,$$

where $\ker(T - \lambda I|_W) = \ker(T - \lambda I) \cap W$ has at least q linearly independent eigenvector (initial vectors). This forces $\dim(W) = k$. ■

Theorem 7.6 .

There exists a basis γ_λ of generalized eigenspace K_λ such that γ_λ is formed by a union of Jordan chains of the eigenvalue λ .

Proof.

We prove by the induction on the dimension of K_λ . Let $n = \dim(K_\lambda)$, $n = 1$, there are nothing to prove. Now suppose $n < k$ such that K_λ admits a basis formed by a union of Jordan chains, then for $n = k$, consider a restriction $T - \lambda I|_{K_\lambda} : K_\lambda \rightarrow K_\lambda$. The proof strategy goes as following:

1. $\ker(T - \lambda I) \supset \{0\}$, by rank-nullity, $\text{im}(T - \lambda I|_{K_\lambda}) \subset K_\lambda$.
2. Claim: Let $u = \text{im}(T - \lambda I|_{K_\lambda})$, u is the generalized eigenspace of $T|_u$ w.r.t eigenvalue λ . From this claim and induction hypothesis, there exist Jordan chains $\beta_1, \beta_2, \dots, \beta_q$ such that their union, γ , forms a basis of u . So $\dim(u) = \text{rank}(T - \lambda I|_{K_\lambda}) = |\gamma|$.
3. For each i , take the initial vector w_i from β_i . By basis extension theorem, the set $\{w_1, w_2, \dots, w_q, u_1, u_2, \dots, u_s\}$ forms a basis in $\ker(T - \lambda I)$. Since $\dim(K_\lambda) = \text{nullity}(T - \lambda I|_{K_\lambda}) + \text{rank}(T - \lambda I|_{K_\lambda}) = \text{nullity}(T - \lambda I) + \dim(u) = q + s + |\gamma|$.
4. On the other hand, $\beta_i \subseteq \text{im}(T - \lambda I|_{K_\lambda}) \subseteq \text{im}(T - \lambda I)$, for each i , there exists a pre-image v_i of the end vector of β_i . Let $\tilde{\beta}_i = \beta_i \cup \{v_i\}$ and the set $\tilde{\gamma} = \tilde{\beta}_1 \cup \tilde{\beta}_2 \cup \dots \cup \tilde{\beta}_q \cup \{u_1\} \cup \{u_2\} \cup \dots \cup \{u_s\}$. The initial vectors are linearly independent, so $\tilde{\gamma}$ is linearly independent, and $|\tilde{\gamma}| = |\gamma| + q + s$. Hence, $\tilde{\gamma}$ forms a maximal linearly independent set in K_λ and hence a basis.

■

Theorem 7.7 .

Let λ and μ be distinct eigenvalues of $T : V \rightarrow V$.

1. K_λ is a T -invariant subspace.
2. The restriction of $T - \mu I$ to K_λ is invertible.

Proof.

For 1, use T and I can commute. Given that $T - \mu I|_{K_\lambda}$ is well-defined by 1, we want to show this restriction is bijective. By TFAE, taking any vector $\mathbf{v} \in \ker(T - \mu I) \cap K_\lambda$, we hope $\mathbf{v} = \mathbf{0}$. Suppose it does not. Since $\mathbf{v} \in K_\lambda$, there exists a smallest integer p s.t. $(T - \lambda I)^p \mathbf{v} = \mathbf{0}$. Let $\mathbf{y} = (T - \lambda I)^{p-1} \mathbf{v}$ such that $T\mathbf{y} = \lambda \mathbf{y}$. On the other hand, $\mathbf{v} \in \ker(T - \mu I)$, we can show that

$\mathbf{y} \in \text{Ker}(T - \mu I)$, using commutativity btw T and I . Then $T\mathbf{y} = \mu\mathbf{y}$, which yields a contradiction. ■

Theorem 7.8 .

Let a be the algebraic multiplicity of the eigenvalue λ of T , then we have $K_\lambda = \text{Ker}(T - \lambda I)^a$.

Proof.

$(\supseteq)$ is clear, for $(\subseteq)$, take $\mathbf{v} \in K_\lambda$, write $p_T(x) = (x - \lambda)^a(x - \mu_1)^{b_1}(x - \mu_2)^{b_2} \cdots (x - \mu_k)^{b_k}$, where $\{\lambda, \mu_1, \mu_2, \dots, \mu_k\}$ are distinct. By Cayley-Hamilton, one has: $p_T(T)(\mathbf{v}) = (T - \lambda I)^a(T - \mu_1 I)^{b_1}(T - \mu_2 I)^{b_2} \cdots (T - \mu_k I)^{b_k}(\mathbf{v}) = \mathbf{0}(\mathbf{v}) = \mathbf{0}$. Consider the map $(T - \mu_1 I)^{b_1}(T - \mu_2 I)^{b_2} \cdots (T - \mu_k I)^{b_k}$ restricted to K_λ , which is invertible by theorem (7.7), then $(T - \lambda I)^a(\mathbf{v}) = \mathbf{0}$. This completes the proof. ■

Remark .

Recall theorem (7.4), the above theorem asserts that $m \leq a$.

Theorem 7.9 .

Let λ and μ be distinct eigenvalues of T , then $K_\lambda \cap K_\mu = \{\mathbf{0}\}$.

Proof.

Take $\mathbf{v} \in K_\lambda \cap K_\mu$ and $p_T(x) = (x - \lambda)^a(x - \mu)^b q(x)$. Since $\gcd\{(x - \lambda)^a, (x - \mu)^b\} = 1$, by Euclidean algorithm, there exists f, g s.t. $f(x)(x - \lambda)^a + g(x)(x - \mu)^b = 1$. Plug $x = T$ and apply the equation to $\mathbf{v}$, we must have $\mathbf{v} = \mathbf{0}$ by theorem (7.8) ■

Theorem 7.10 (Primary Decomposition Theorem).

Let V be a finite-dimensional vector space over $\mathbb{C}$ and $T : V \rightarrow V$ be a linear operator.

Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of T , then:

$$V = K_{\lambda_1} \oplus K_{\lambda_2} \oplus \cdots K_{\lambda_k}.$$

Proof.

Let $U = K_{\lambda_1} \oplus K_{\lambda_2} \oplus \cdots K_{\lambda_k}$. We hope $\dim(U) = \dim(V)$. (Direct) sum of space is a subspace, $U \subseteq V$ implies $\dim(U) \leq \dim(V)$. Next, $\dim(U) = \sum_{i=1}^k \dim(K_{\lambda_i}) \geq \sum_{i=1}^k \text{nullity}(T - \lambda_i I)^{a_i} \geq \text{nullity}(\prod_{i=1}^k (T - \lambda_i I)^{a_i}) = \text{nullity}(\mathbf{0}) = k$. Combine $U \subseteq V$ and $\dim(U) = \dim(V)$, we conclude

$$U = V. \quad \blacksquare$$

Theorem 7.11 (JCF Theorem I).

Let V be a finite dimensional vector space over complex field, $T : V \rightarrow V$ be any linear operator, and the corresponding matrix representing T w.r.t standard basis, $[T]_B^B$, then there exists a basis γ of V formed by a union of Jordan chains, which is called a Jordan basis. Moreover, $[T]_B^B$ similar to its JCF, $[T]_\gamma^\gamma$.

Proof.

By primary decomposition theorem (7.10) and theorem (7.6), there exists such union of Jordan chains and forms a maximal linearly independent set and hence a basis. $\blacksquare$

Theorem 7.12 (JCF Theorem II & III).

Let $A, B \in M_n(\mathbf{C})$, and up to a permutation of Jordan blocks, then:

1. J_A is uniquely determined by A .
2. $A \sim B$ if and only if $J_A = J_B$.

Proof.

1 is trivial by definition. For 2, we focus on $(\Rightarrow)$. By similarity, $\text{Spec}(A) = \text{Spec}(B)$ and $(A - \lambda I)^i \sim (B - \lambda I)^i$, which implies $\text{nullity}(A - \lambda I)^i = \text{nullity}(B - \lambda I)^i$ for each integer i . Hence, they have the same JCF. $\blacksquare$

8 Inner Product Space

Definition 8.1 (Inner Product Space).

Let V be a vector space over $\mathbb{C}$ and $\langle -, - \rangle : V \times V \rightarrow \mathbb{C}$ be a function. We say the pair $(V, \langle -, - \rangle)$ is a complex inner product space if it satisfies:

1. (Linearity in 1st component) For any $u, v, w \in V$ and $\alpha, \beta \in \mathbb{C}$:

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle.$$

2. (Conjugate Symmetry) For any $u, v \in V$:

$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$

3. (Non-negativity) For any $u \in V$:

$$\langle u, u \rangle \geq 0.$$

4. (Non-degeneracy) For any $u \in V$:

$$\|u\| = \sqrt{\langle u, u \rangle} = 0$$

iff $u = 0$.

And $\langle -, - \rangle$ is called an inner product on V .

Remark .

Dot product is the standard inner product on $\mathbb{C}^n$.

Definition 8.2 .

Let $(V, \langle -, - \rangle)$ be a complex inner product space.

1. The norm of $x \in V$ is defined as $\|x\| = \sqrt{\langle x, x \rangle}$.
2. $x, y \in V$ are orthogonal if $\langle x, y \rangle = 0$.

Theorem 8.1 .

Let $(V, \langle -, - \rangle)$ be a complex inner product space, for $u, v, w \in V$ and $\alpha, \beta \in \mathbb{C}$.

1. Almost linear in 2nd comp: $\langle u, \alpha v + \beta w \rangle = \bar{\alpha} \langle u, v \rangle + \bar{\beta} \langle u, w \rangle$.
2. $\|\alpha u\| = |\alpha| \|u\|$.
3. $\langle u, v \rangle = 0$ for any $v \in V \Leftrightarrow u \equiv 0$.
4. $\langle u, w \rangle = \langle v, w \rangle$ for any $w \in V \Leftrightarrow u = v$.

Proof.

Left as an exercise. ■

Theorem 8.2 .

Let $(V, \langle -, - \rangle)$ be a complex inner product space, for $u, v \in V$.

- **Cauchy-Schwarz:** $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$.
- **Triangle:** $\|u + v\| \leq \|u\| + \|v\|$.

Proof.

For Cauchy-Schwarz, if $v = 0$, then it is trivial. Assume $v \neq 0$, inspired by the projection vector in $\mathbb{R}^2$, consider $z = u - \frac{\langle u, v \rangle}{\|v\|^2} v$. First show z, v are orthogonal, and by non-negativity, $\langle z, z \rangle \geq 0$. After some expansion, we have

$$\|u\|^2 - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, u \rangle \geq 0,$$

rearrange and take the square root, then we are done.

For triangle inequality,

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \langle u, u \rangle + 2\operatorname{Re}(\langle u, v \rangle) + \langle v, v \rangle \\ &\leq \|u\|^2 + 2|\langle u, v \rangle| + \|v\|^2 \\ &\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \end{aligned}$$

Hence $\|u + v\| \leq \|u\| + \|v\|$. ■

Theorem 8.3 .

Let $(V, \langle -, - \rangle)$ be a complex inner product space. For any $u, v \in V$, we have:

- **Parallelogram Identity** $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$.
- **Generalized Pythagorean Theorem** If $u_1, u_2, \dots, u_n$ are mutually orthogonal (i.e., $\langle u_i, u_j \rangle = 0$ for all $i \neq j$), then

$$\left\| \sum_{i=1}^n u_i \right\|^2 = \sum_{i=1}^n \|u_i\|^2.$$

- **Polarization Identity**

$$\langle u, v \rangle = \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 + i\|u + iv\|^2 - i\|u - iv\|^2 \right).$$

Proof.

Left as an exercise. ■

Theorem 8.4 (Jordan-von Neumann).

Let V be a vector space equipped with a norm $\|\cdot\|$. This norm arises from an inner product (i.e., there exists an inner product on V such that $\|v\| = \sqrt{\langle v, v \rangle}$ for all $v \in V$) if and only if the norm satisfies the parallelogram identity:

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2) \quad \forall u, v \in V.$$

Proof.

Beyond the scope of this note. ■

Definition 8.3 (Orthogonal and Orthonormal Sets).

Let $(V, \langle -, - \rangle)$ be an inner product space. A subset $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of V is called an **orthogonal set** if every pair of distinct vectors is orthogonal, i.e.,

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \quad \text{for all } i \neq j.$$

Furthermore, S is called an **orthonormal set** if it is an orthogonal set and every vector in S is a unit vector. Compactly, we can write the condition for an orthonormal set using the Kronecker delta δ_{ij} :

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Theorem 8.5 .

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal set of non-zero vectors in an inner product space V . Then S is linearly independent.

Theorem 8.6 .

Let $(V, \langle -, - \rangle)$ be a finite-dimensional complex inner product space. If $S = \{u_1, u_2, \dots, u_k\}$ is an orthogonal basis of V , then for any $v \in V$,

$$v = \sum_{i=1}^k \alpha_i u_i, \quad \text{where } \alpha_i = \frac{\langle v, u_i \rangle}{\|u_i\|^2}.$$

In particular, if S is an ONB, then $\alpha_i = \langle v, u_i \rangle$.

Theorem 8.7 .

Let $(V, \langle -, - \rangle)$ be a finite-dimensional complex inner product space. Let $\beta = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be an orthonormal basis (ONB) for V . For any $\mathbf{v}, \mathbf{w} \in V$, let $[\mathbf{v}]_\beta$ and $[\mathbf{w}]_\beta$ be their coordinate vectors relative to β . Then the inner product of $\mathbf{v}$ and $\mathbf{w}$ is equal to the standard

dot product of their coordinate vectors:

$$\langle \mathbf{v}, \mathbf{w} \rangle_V = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\text{dot}}$$

Theorem 8.8 (Gram-Schmidt Process).

Let $(V, \langle -, - \rangle)$ be an inner product space. Given a linearly independent set $S = \{v_1, v_2, \dots, v_l\}$ in V , if

1. $w_1 = v_1$,
2. $w_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, w_j \rangle}{\|w_j\|^2} w_j$ for $2 \leq k \leq l$.

Then:

1. $S' = \{w_1, w_2, \dots, w_l\}$ is an orthogonal set of non-zero vectors.
2. $\text{Span}(S) = \text{Span}(S')$.

Proof.

We shall prove by induction on $l = |S|$. For $l = 1$, there is nothing to prove. Suppose the claim is true for $l = k - 1$, then for $l = k$, we need to check:

1. $w_k \neq 0$.
2. $\langle w_k, w_i \rangle = 0$ for $1 \leq i \leq k-1$.
3. Take

$$w_k \in \text{Span}(\{w_1, w_2, \dots, w_{k-1}, v_k\}) = \text{Span}(\{v_1, v_2, \dots, v_{k-1}, v_k\}),$$

so $\text{Span}(S') \subseteq \text{Span}(S)$. Note that $\text{Span}(S)$ is given to be linearly independent and $\text{Span}(S')$ is linearly independent by orthogonality of non-zero vectors, so both sets have the same dimension. Hence $\text{Span}(S') = \text{Span}(S)$. ■

Definition 8.4 (Orthogonal Projection).

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V . Let $S = \{u_1, u_2, \dots, u_k\}$ be an ONB of E . The projection operator onto E is defined as:

$$P_E(v) = \sum_{j=1}^k \langle v, u_j \rangle u_j.$$

Remark .

1. The projection operator $P_E : V \rightarrow E \subseteq V$ is a linear operator.
2. P_E is an idempotent operator, i.e. $P_E^2 v = P_E v \quad \forall v \in V$.

Theorem 8.9 .

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and $P_E : E \rightarrow V$ be the projection onto a subspace E . Then:

1. $P_E(v) \in E$.
2. $v - P_E(v)$ is orthogonal to every vector in E .

Proof.

For 1, it is clear by definition.

For 2, it suffices to check $\langle v - P_E(v), u_j \rangle = 0 \quad \forall j = 1, 2, \dots, k$. ■

Theorem 8.10 (Minimal Distance Property).

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V . Fix any $v \in V$, then:

1. Among all vectors in E ,

$$\|P_E(v) - v\| \leq \|z - v\| \quad \forall z \in E.$$

2. If the equality holds, then $z = P_E(v)$.

Proof.

For 1, observe that $\langle z - P_E(v), v - P_E(v) \rangle = 0$, then we can use Pythagorean theorem.

For 2, the equality holds means $\|z - P_E(v)\| = 0$. By non-degeneracy, $z = P_E(v)$. ■

Definition 8.5 (Orthogonal Complement).

Let $(V, \langle -, - \rangle)$ be an inner product space and E be a subset of V . The orthogonal complement of E is defined as:

$$E^\perp = \{v \in V : \langle v, u \rangle = 0 \quad \forall u \in E\}.$$

Remark .

1. E is not necessarily a subspace.
2. $E^\perp$ is a subspace of V .

Theorem 8.11 .

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V .

1. $v - P_E(v) \in E^\perp$.
2. $\text{Ker}(P_E) = E^\perp$.

Proof.

For 1, it is followed by $v - P_E(v)$ is orthogonal to any vector in E .

For 2, take $v \in \text{Ker}(P_E)$, then $v - P_E(v) = v \in E^\perp$. Conversely, take $v \in E^\perp$, then $P_E(v) = \underbrace{v - P_E(v)}_{\in E^\perp} + v = P_E(v) \in E^\perp$.

By non-degeneracy, $P_E(v) = 0$. ■

Theorem 8.12 (Orthogonal Decomposition).

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V .

1. $V = E \oplus E^\perp$
2. $\dim(V) = \dim(E) + \dim(E^\perp)$.

Proof.

For 1, observe that $v = P_E(v) + (v - P_E(v))$ for any $v \in V$. To show E and $E^\perp$ has trivial intersection, we take $v \in E \cap E^\perp$, then $\langle v, v \rangle = 0$. By non-degeneracy, $v = 0$.

For 2, it follows from 1 directly. ■

Theorem 8.13 .

Let $A \in M_{m \times n}(\mathbb{R})$, $x \in \mathbb{R}^n$ $y \in \mathbb{R}^m$.

1. $\langle Ax, y \rangle_{\text{dot}} = \langle x, A^\top y \rangle_{\text{dot}}$.
2. $\text{rank}(A^\top A) = \text{rank}(A)$.
3. If $\text{rank}(A) = n$, then $A^\top A$ is invertible.

Proof.

For 1 and 3, it is an easy exercise.

For 2, we first show $\text{ker}(A^\top A) = \text{ker}(A)$.

For any $v \in \ker(A^\top A)$, $\langle A^\top A v, v \rangle = 0$. By 1, $\langle A v, A v \rangle = 0$, so $A v = 0$ by degeneracy. The converse is trivial. In addition, both A and $A^\top A$ shares the same dimension of domain (The matrices have the same number of columns). By rank-nullity, $\text{rank}(A^\top A) = \text{rank}(A)$. ■

Question (Best Solution Problem).

Given a system of equations $X\beta = y$ to be inconsistent, where $X \in M_{m \times n}(\mathbb{R})$, $y \in \mathbb{R}^m$. Find $\beta_0 \in \mathbb{R}^n$ that minimizes $\|X\beta - y\|$, i.e.

$$\|X\beta_0 - y\| \leq \|X\beta - y\|$$

for any $\beta \in \mathbb{R}^n$.

Remark .

The norm is induced by the usual dot product on real vector space.

To solve the best solution problem, we employ the following method:

Theorem 8.14 (Least Square Methods).

Let $X \in M_{m \times n}(\mathbb{R})$, $y \in \mathbb{R}^m$. Suppose $\beta = \beta_0 \in \mathbb{R}^n$ is a minimizer of $\|y - X\beta\|$, then:

1. β_0 satisfies $X^\top X\beta = X^\top y$;
2. If $\text{rank}(X) = n$, then $\beta_0 = (X^\top X)^{-1} X^\top y$.

Proof.

For 1, let $W = \text{Im}(X) = \{X\beta : \beta \in \mathbb{R}^n\}$. By the minimal distance property,

$$\begin{aligned} \|P_W(y) - y\| &\leq \|w - y\| \quad \forall w \in W \\ \Rightarrow \|P_W(y) - y\| &\leq \|X\beta - y\| \quad \forall \beta \in \mathbb{R}^n. \end{aligned}$$

So it suffices to find β_0 such that $P_W(y) = X\beta_0$. This is possible since $P_W(y) \in W$. Let $P_W(y) = X\beta_0$ for some $\beta_0 \in \mathbb{R}^n$. Note that

$$\langle P_W(y) - y, w \rangle = \langle X\beta_0 - y, X\beta \rangle = \langle X^\top (X\beta_0 - y), \beta \rangle = 0 \quad \forall \beta \in \mathbb{R}^n,$$

then we have $X^\top (X\beta_0 - y) = 0$.

For 2, it is an easy exercise. ■

Question (Minimal Solution Problem).

Given a system of equations $Ax = b$ to be infinitely many solutions, where $A \in M_{m \times n}(\mathbb{R})$, $b \in \mathbb{R}^m$. Find $u \in \mathbb{R}^n$ that s.t. $\|u\| \leq \|x\|$ for all solutions x to the system.

Theorem 8.15 .

Let $A \in M_{m \times n}(\mathbb{R})$, then $\text{Ker}(A) = (\text{Im}(A^\top))^\perp$.

Proof.

Left as an exercise. ■

Theorem 8.16 .

Given a system of equations $Ax = b$ to be infinitely many solutions, where $A \in M_{m \times n}(\mathbb{R})$, $b \in \mathbb{R}^m$.

1. There exists a unique minimal solution u to the system and $u \in \text{Im}(A^\top)$.
2. Let v be a solution to $AA^\top v = b$, then $A^\top v = u$.

Proof.

For 1, for any solution x_0 to the system, we can write a general solution $u = x_0 - v$ for some $v \in \text{Ker}(A)$. It suffices to find v such that $\|x_0 - v\| \leq \|x_0 - w\|$ for all $w \in \text{Ker}(A)$. Consider the projection $v = P_{\text{Ker}(A)}(x_0)$. By the minimal distance property, v is unique, so $u = x_0 - v \in (\text{Ker}(A))^\perp = \text{Im}(A^\top)$ is also unique.

For 2, suppose $AA^\top v = b$ and $Au = b$ by construction, then $A^\top v - u \in \text{Ker}(A) \cap \text{Im}(A^\top) = \{0\}$. Hence $A^\top v = u$. ■

Remark .

Why $AA^\top v = b$ is consistent? Note that $\text{Ker}(A^\top) = \text{Ker}(AA^\top)$. Use $\text{Ker}(A) = \text{Im}(A^\top)^\perp$ and take orthogonal complement on both sides, one has: $\text{Im}(AA^\top) = \text{Im}(A)$, then we are done.

Theorem 8.17 .

Let $A \in M_n(\mathbb{C})$ and $v, w \in \mathbb{C}^n$, then

$$\langle Av, w \rangle = \langle v, \overline{A^\top w} \rangle.$$

Definition 8.6 (Adjoint Operator).

Let $A \in M_n(\mathbb{C})$. The adjoint of A is defined by its conjugate transpose, denoted by $A^* = \overline{A^\top}$.

In general, the adjoint of a linear operator T is an operator $T^* : V \rightarrow V$ if

$$\langle Tv, w \rangle = \langle v, T^*w \rangle \quad \forall v, w \in V$$

Theorem 8.18 .

For each linear operator, its adjoint operator exists and is unique.

Proof.

For existence, we first fix an ONB β , and define $[T^*]_\beta^\beta = \overline{([T]_\beta^\beta)^\top}$. Use theorem (8.7), we check that

$$\langle Tv, w \rangle = \langle [Tv]_\beta, [w]_\beta \rangle = \langle [v]_\beta, [T^*]_\beta^\beta [w]_\beta \rangle = \langle v, T^*w \rangle.$$

For uniqueness, we left as an exercise. ■

Theorem 8.19 .

Let $T, U : V \rightarrow V$ be two linear operators on an inner product space V . We have:

1. $(T + U)^* = T^* + U^*$.
2. $(cT)^* = \bar{c}T^*$ for any $c \in \mathbb{C}$.
3. $(TU)^* = U^*T^*$.
4. $(T^*)^* = T$.

Theorem 8.20 .

Let S be an ONB of V , and S' be another basis of V . TFAE:

1. S' is also an ONB of V .
2. The change of coordinate matrix U from S' to S satisfies $U^*U = U^*U = I_n$ (i.e. U is unitary).

Definition 8.7 .

Let $A \in M_n(\mathbb{C})$,

1. A is normal if $AA^* = A^*A$.
2. A is unitary if $AA^* = A^*A = I_n$ (i.e. $A^* = A^{-1}$).
3. A is Hermitian/self-adjoint if $A = A^*$.

Theorem 8.21 .

For a finite-dimensional inner product space V , the set of unitary operators forms a group. That is,

$$U(V) = \{T : V \rightarrow V : T \text{ is unitary}\} \text{ is}$$

1. closed under composition,
2. closed under inversion.

Theorem 8.22 .

Let $T \in U(V)$ and $\lambda \in \text{Spec}(T)$. Then $\|\lambda\| = 1$, i.e.

$$\lambda = e^{i\theta} = \cos \theta + i \sin \theta \quad \forall \theta \in \mathbb{R}.$$

Proof.

Compute $\langle Tv, Tv \rangle = \langle v, T^*Tv \rangle = \langle v, v \rangle$ by unitarity of T . On the other hand, $\langle Tv, Tv \rangle = \langle \lambda v, \lambda v \rangle = \lambda \bar{\lambda} \|v\|^2$. Compare them and $v \neq 0$, we obtain $|\lambda| = 1$. ■

Theorem 8.23 .

Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and $T : V \rightarrow V$ be a linear operator. Then T is a unitary operator if and only if T is an isometry. That is,

$$T \in U(V) \iff \|T\mathbf{v}\| = \|\mathbf{v}\| \quad \forall \mathbf{v} \in V.$$